Security challenges in datacenter networks: Risks and Mitigation strategies

Stephen Mc Neill Killeen

Dr. Kevin Murphy

Bachelor of Science in Cyber Security and IT Forensics
Spring semester 2026 Final Presentation

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Presentation overview

- Project Overview
- Project Implementation
- Key Results
- Problems and solutions
- Future Work
- Demonstration
- Conclusion

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Project overview – Introduction

- Modern datacenter networks generate large volumes of distributed network traffic, making security monitoring increasingly complex

- Traditional monitoring approaches struggle to analyse large-scale, multi-source log data efficiently

- This project develops a Python-based log analysis system to simulate automated security monitoring

- The system processes firewall, DNS, IDS, and authentication logs to detect suspicious behaviour

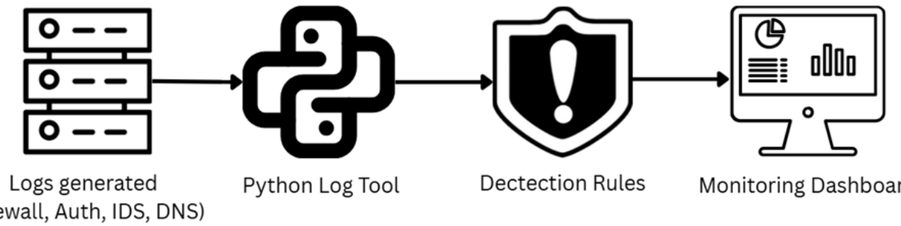
- The aim is to demonstrate how automated log analysis improves visibility and threat detection

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Project overview - Functionality

- Simulates a datacenter monitoring environment
- Aggregates logs from multiple distributed sources
- Applies rule-based detection techniques
- Identifies suspicious patterns across network activity
- Visualises results using a monitoring dashboard

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System Architecture

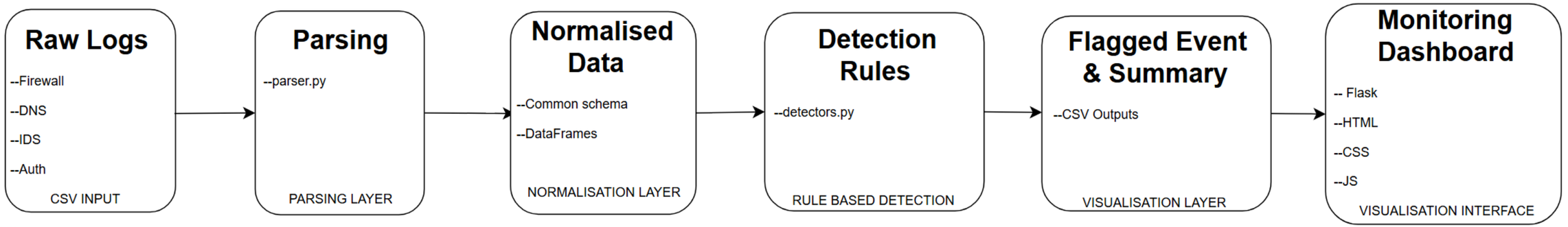
Project implementation - Architecture

- The system is designed as a modular log analysis architecture
- Multiple log sources are simulated, including firewall, DNS, IDS, and authentication systems
- Logs are collected and processed centrally by the Python analysis engine
- This approach allows correlation of events across different network components
- The analysed results are then visualised through a monitoring dashboard

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Project implementation – Log Processing

- Raw log data is generated and stored in text-based formats
- A parsing layer extracts key attributes such as timestamps, IP addresses, and event types
- Logs are converted into structured formats (e.g. DataFrames) for efficient analysis
- Data is normalised into a common schema to handle different log formats
- This enables consistent processing across all log sources



Log Processing Pipeline

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Project implementation - Detection

- Detection is performed using rule-based analysis of log data
- The system identifies patterns that deviate from expected behaviour
- Examples of detection rules include:
  - Repeated firewall deny events (potential scanning)
  - Multiple failed login attempts (brute-force attacks)
  - Unusual DNS query patterns
  - IDS alerts indicating possible intrusion
- Detection uses configurable thresholds and time windows to improve accuracy

Example detection rule implementation

```
#Detect repeated firewall denies
flagged_denies = detect_repeated_denies(
    df_firewall,
    window_minutes=w_brute,
    threshold=fw_threshold_denies
)
```

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Key results

- The system successfully detected multiple types of suspicious behaviour
- Repeated firewall deny events indicated potential scanning activity
- Multiple failed login attempts identified possible brute-force attacks
- Unusual DNS activity highlighted abnormal network behaviour
- IDS alerts were captured and correlated with other log sources
- Combining multiple log sources improved detection accuracy and visibility

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Key results



Monitoring Dashboard Output

- Real-time alert overview
- Visualisation of log activity trends
- Breakdown of alert types
- Supports quick identification of anomalies

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Problems and solutions

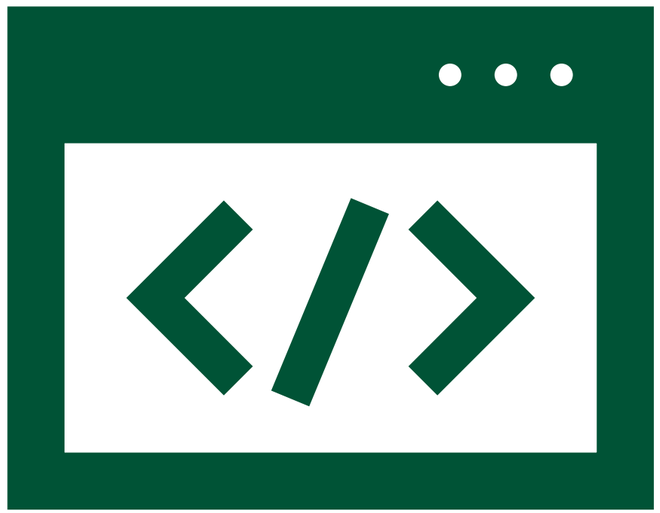
PROBLEMS

- High volume of distributed log data
- Inconsistent log formats
- Manual analysis inefficiency
- Risk of false positives

SOLUTIONS

- Automated log processing pipeline
- Parsing and data normalisation
- Rule-based detection engine
- Threshold tuning and filtering

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Potential uses of the system and future work

Uses:

- Lightweight log monitoring solution
- Supports analysis of network activity
- Demonstrates SIEM like functionality
- Can integrate with enterprise systems

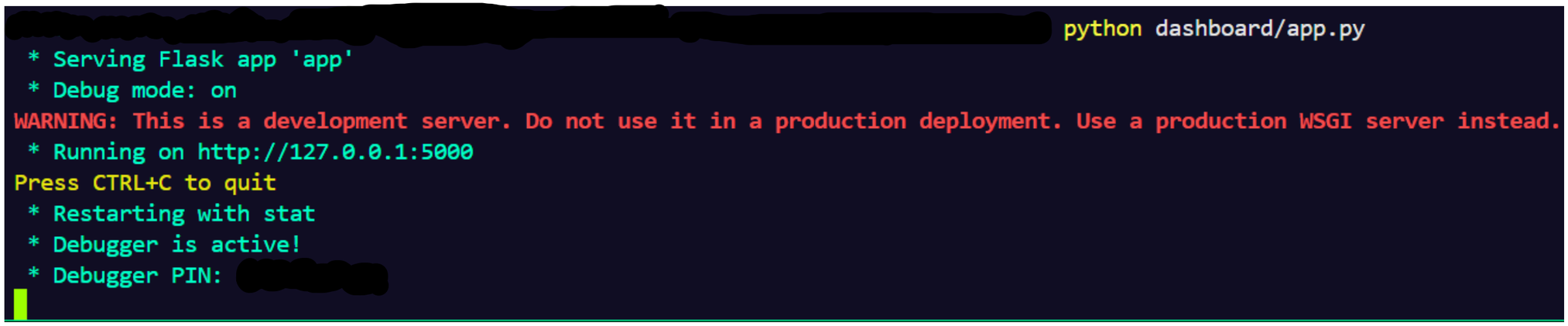
Future Work:

- Real-time log processing
- Machine learning / anomaly detection
- Additional log sources (cloud, endpoints)
- Improved dashboard visualisation

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Demonstration

- Generate simulated log data (firewall, DNS, IDS, authentication)
- Execute the Python log analysis system
- Apply detection rules to identify suspicious behaviour
- Display flagged events and summaries in the monitoring dashboard

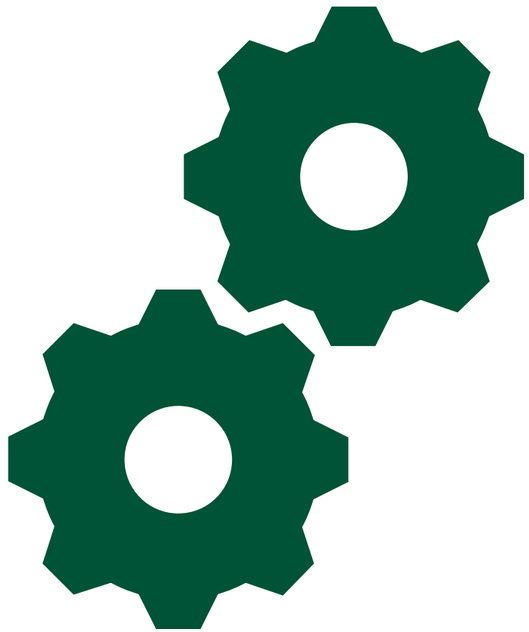


System execution output

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Conclusions

- Datacenter environments produce large volumes of distributed log data
- Automated log analysis improves visibility and reduces manual effort
- Rule-based detection effectively identifies common attack patterns
- The developed system demonstrates a practical approach to network monitoring
- Future improvements can enhance scalability and detection capabilities



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Thank you.
Any questions?

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University of Limerick,
Limerick, V94 T9PX,
Ireland.

Ollscoil Luimnigh,
Luimneach,
V94 T9PX, Éire.

+353 (0) 61 202020

ul.ie